

Reclaiming the writing process using AI: a scaffolded writing assignment that promotes critical thinking, reflection, and accountability

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ABSTRACT As the educational landscape changes with the widespread use of generative artificial intelligence (AI), we have made the deliberate choice to adapt with intention through the constructive integration of this tool within the curriculum. Writing assignments are often vulnerable to AI shortcuts, and therefore, we reimagined the project in our online microbiology lab to incorporate AI in a structured and pedagogically sound way that prioritizes both the writing process and the final product. Students must work with AI to create a script for a 5-minute video which they then illustrate and record on the differences between *Staphylococcus aureus* and methicillin-resistant *S. aureus*. The drafting process is highly structured and provides a scaffold for students to reflect on their own knowledge, think critically about AI outputs, develop AI literacy skills, and engage in self-reflection. Students are graded based on their involvement in this modified writing process rather than the final product. Subsequently, students create illustrations and record their 5-minute video; engaging in this verbal and visual communication helps reinforce student learning. Surveys conducted in the Fall 2025 semester suggest that the assignment reinforced student content understanding and reiterated the importance of responsible AI use. Additionally, students self-reported increased confidence in their prompt engineering skills. This assignment achieves student engagement and fosters the growth of AI literacy without compromising critical thinking or scholarly diligence. It can be adapted to different fields and topics and showcases a purposeful, strategic, and thoughtful incorporation of AI in undergraduate education.

KEYWORDS writing assignment, AI literacy, critical thinking, student reflection, AI integration

With the widespread use of generative artificial intelligence (AI), many of our traditional assignments are susceptible to being bypassed when students turn to ChatGPT and other AI platforms (1–3). The conventional lab report formerly assigned in our online asynchronous microbiology course suffered from a lack of student enthusiasm and returned a notable number of apparently AI-assisted submissions. Furthermore, students did not have to engage in extensive critical thinking, effective prompt engineering, or deep analysis of the AI output to arrive at a passable written product. Due to the limitations of this traditional assignment, students could easily bypass the learning process, missing essential steps for retention and deeper understanding.

Concerns about student disengagement and dependence on AI have fueled the rise of advocates for change in teaching practices. Educators are pressed to reimagine assignments to foster deeper engagement, critical thinking, and meaningful learning in an AI-integrated academic environment, rather than focusing on enhanced detection tools, implementing restrictive measures, or viewing students as dishonest (4).

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Many publications have addressed the myriad opportunities available to educators as they seek to integrate AI across the curriculum (5–7). However, there are concomitant concerns with widespread integration of this technology, namely, that it negatively impacts academic performance (8), promotes overreliance and the loss of foundational skills (9), and can drastically impact our students' analytical thinking (6, 10, 11). Yet the undeniable fact remains that AI is here to stay. How do we reconcile the potential AI has to offer with our pedagogical concerns?

In this paper, we illustrate how students navigate a two-part assignment in which they work independently with AI to create a 5-minute video where they juxtapose *Staphylococcus aureus* and methicillin-resistant *S. aureus* and account for their differential susceptibility to penicillin and its derivatives. In Part 1, students engage in a modified writing process to draft and revise the script for their 5-minute video. By scaffolding an assignment into its constituent parts, faculty can monitor students and gain deeper insight into their thinking (12, 13). Producing their script enables students to develop AI literacy skills, think critically about AI outputs, and engage in self-reflection. In Part 2, students are tasked with illustrating and recording their 5-minute video without AI assistance. The intent of Part 2 is to hold students accountable for their understanding of the information within their script and enable them to demonstrate their knowledge through illustrations. Our innovative approach to this writing assignment balances AI integration into the writing process without sacrificing critical thinking or learning outcomes on the part of our students.

Intended audience

This assignment is intended for undergraduate microbiology students, both majors and non-majors, though it is adaptable to other contexts. It is most suited for topics that integrate technical content with broader conceptual understanding. Assignments lacking sufficient technical detail at the undergraduate level may be less appropriate as the illustrative component in Part 2 is essential for ensuring student accountability and learning (Appendix 1).

PROCEDURE

In Part 1 of the assignment, students are introduced to an iterative scripting process aided by AI, leading to the creation of a 5-minute video script. A worksheet helps students navigate Part 1 independently, and the submitted work allows for instructor assessment (Fig. 1A).

To begin, we asked students to analyze their current knowledge, which establishes a foundation for their script. To facilitate this, we developed a framework to engage students in structured self-reflection. The CLARITY framework (context, learning goal, audience, requirements, information check, tone, your request) requires students to articulate their thoughts on paper, thereby making their reasoning visible (Table 1). To promote AI literacy skills, students are subsequently asked to write their initial prompt in which they direct the AI of their choice to draft their script; completion of the CLARITY framework provides them with the necessary information to effectively accomplish this task (14). Students are instructed to include, in their initial prompt, a request that the chatbot pose 5 clarifying questions before generating the first draft of the script. These questions are designed to help identify missing information, reveal assumptions, and prompt students to supply essential detail, thereby promoting critical thinking and leveraging AI technology by enabling the student to receive a personalized set of questions depending on their initial prompt (7, 15). The students have to think critically and record their responses before continuing.

After completing these steps, students begin to engage with the script produced by AI. To ensure that students are meeting a required minimum number of interactions with the AI and evaluating the output, they must provide evidence of iterative refinement. In order to track the student's progress and provide guidance, we created the REACT log (review, evaluate, adjust, clarify, try again) (Table 2). This table helps students to

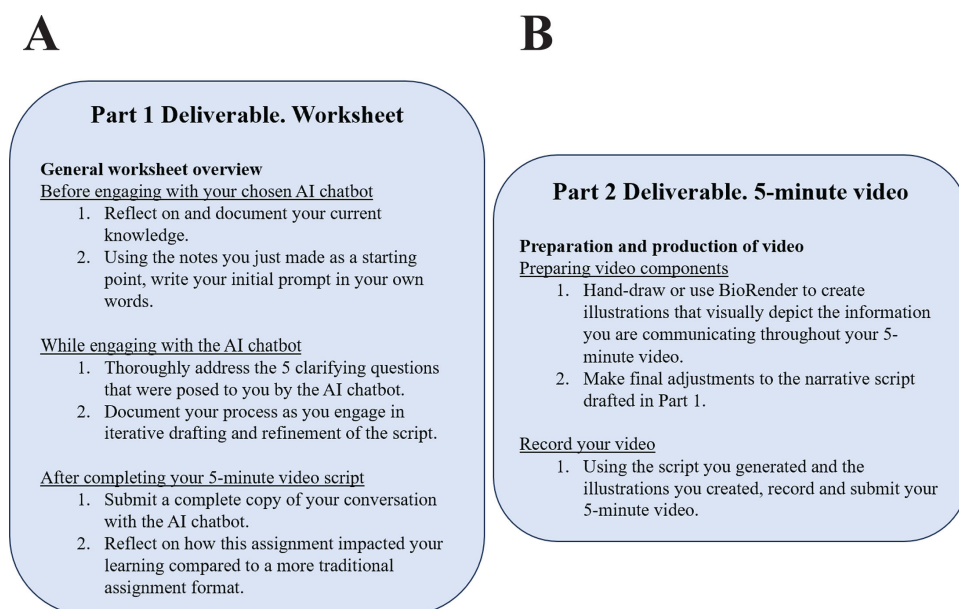


FIG 1 Overview of the writing assignment. (A) A general overview of the worksheet students complete as they progress independently through Part 1 of the writing assignment. (B) Review of the steps taken by students as they complete Part 2 of the writing assignment.

engage in the critical act of analysis and evaluation of the AI output as well as organize their thoughts in preparation for writing their next prompt (16). Lastly, students answer a self-reflection question (Fig. 1A), a practice shown to enhance student learning and metacognition (17–19). Part 1 is augmented through the integration of AI and does not compromise student engagement in the iterative writing process. Emphasis on the process over the product is evident in the questions students must answer throughout the assignment; the grading rubric for Part 1 reflects our goals by rewarding thorough completion of the deliverable rather than the actual script itself (Appendix 2).

In Part 2, this assignment holds students accountable for their understanding of the written material (Fig. 1B). First, students must create illustrations that will accompany the narration they just developed. To promote student engagement and learning, they must illustrate by hand or use icons such as those available in BioRender to construct their visual depictions. Images obtained through an internet search engine, textbook, etc., are prohibited as pedagogical literature suggests that the act of drawing promotes higher-order thinking and memory formation, thereby facilitating student learning (20, 21). Furthermore, requiring students to illustrate their video helps expose them to different

TABLE 1 CLARITY framework

Letter	Category	Reflection questions
C	Context	What background knowledge is important for understanding this topic? What does the viewer need to know to follow this script?
L	Learning goal	What should the viewer understand after watching this video? What's the "big takeaway" I want to communicate?
A	Audience	Who is this script for (e.g., classmates, general public, high school students)? What prior knowledge can I assume they have?
R	Requirements (scope)	What topics do I want the script to include (and avoid)?
I	Information check	What do I already know about this topic? What am I unsure about or need help explaining clearly?
T	Tone	How should this sound? What tone will make the video engaging and understandable?
Y	Your request	What exactly am I asking the AI chatbot to do?

TABLE 2 REACT log

Letter	Step	Reflection questions
R	Review	What worked well in the last draft?
E	Evaluate	What was unclear, inaccurate, or off-target?
A	Adjust	What specific part are you asking the AI chatbot to revise or improve?
C	Clarify	What exact follow-up prompt did you send to the AI chatbot? (copy/paste or summarize)
T	Try again	Did the new response improve the script? What still needs work?

communication mediums. The importance of diverse forms of communication cannot be understated (22), and visual communication has been well documented to improve student learning outcomes (23, 24). Importantly, student illustrations are evaluated based on clarity and thorough attention to technical content and not for artistic quality or drawing skill. Once the illustrations are complete, students have significant autonomy in determining the format of their 5-minute video. Part two focuses on the overall product but nevertheless requires extensive student engagement and holds learners accountable for their own understanding of the material they are presenting (Appendix 3). The learning goals for this assignment map well to representative course/program goals and institutional learning goals (Table 3).

CONCLUSIONS

Students first completed a voluntary and anonymous 6-question pre-survey (Appendix 4) prior to beginning Part 1 of the assignment. Part 1 was supplemented with a grading rubric and two-page guide on effective AI use (Appendix 5). Subsequently, students completed Part 2 of the assignment and 6 post-survey questions (Appendix 6).

All but one of the pre-survey respondents felt confident in their grasp of the material (Fig. 2A), and for many, but not all, it seems that engaging in this assignment reinforced this content understanding (Fig. 2B). In contrast to their self-reported content confidence, pre-survey respondents perceived themselves to be comparatively less prepared to effectively communicate this information (Fig. 2C). However, 95% of post-survey respondents felt that they produced a script that clearly and effectively communicated

TABLE 3 Learning outcomes

Assignment learning goals	Representative	
	Course or program goals	Institutional learning goals
Part 1		
1. Students gain a deep understanding of the differences between <i>S. aureus</i> and MRSA.	Remember, understand, and apply unifying core biological concepts.	Cultivate interdisciplinary knowledge and demonstrate learning in both broad foundational and focused advanced studies.
2. Students analyze and evaluate how these differences directly impact real-world clinical decisions.		
Part 2		
1. Students engage in learning through drawing.		
2. Students are held accountable for their knowledge through the creation of a 5-minute video.		
Part 2	Communicate scientific knowledge in various formats.	Communicate effectively.
1. Students are exposed to diverse communication mediums (written, oral, visual).		
Part 1	Develop AI literacy skills.	Integrate technology into professional practice.
1. Students develop AI literacy through prompt engineering.		
2. Students critically analyze generative AI outputs.		
3. Students review responsible use of AI in an academic setting.		
Part 1	Practice transferrable skills that help develop students into thoughtful learners.	Develop robust learning skills that enable academic success and naturally foster greater ownership of the learning process.
1. Students develop metacognitive skills through self-reflection at the beginning and end of the modified writing process.		

the scientific information (Fig. 2D). Notably, this perception may have been influenced by AI support in framing concepts and supplying an initial draft for revision.

For most, it seems our assignment reiterates previous instruction on the responsible use of AI in an academic setting (Fig. 3A and B). Regarding prompt engineering, 26% of the pre-survey respondents self-identified no strong understanding of prompt engineering strategies to guide their interaction with AI (Fig. 3C). After completing the assignment, only 8% of post-survey respondents believed their understanding and execution of prompt engineering strategies had not improved (Fig. 3D).

Student responses to the self-reflection question posed at the end of Part 1 (Fig. 1A) suggest they found value in actively reviewing the AI output for clarity and accuracy and that the iterative nature of the assignment helped reinforce their learning and identify content in their narrative that was missing or misunderstood. Some students felt that completion of the REACT log was excessive. Regarding AI literacy, student comments

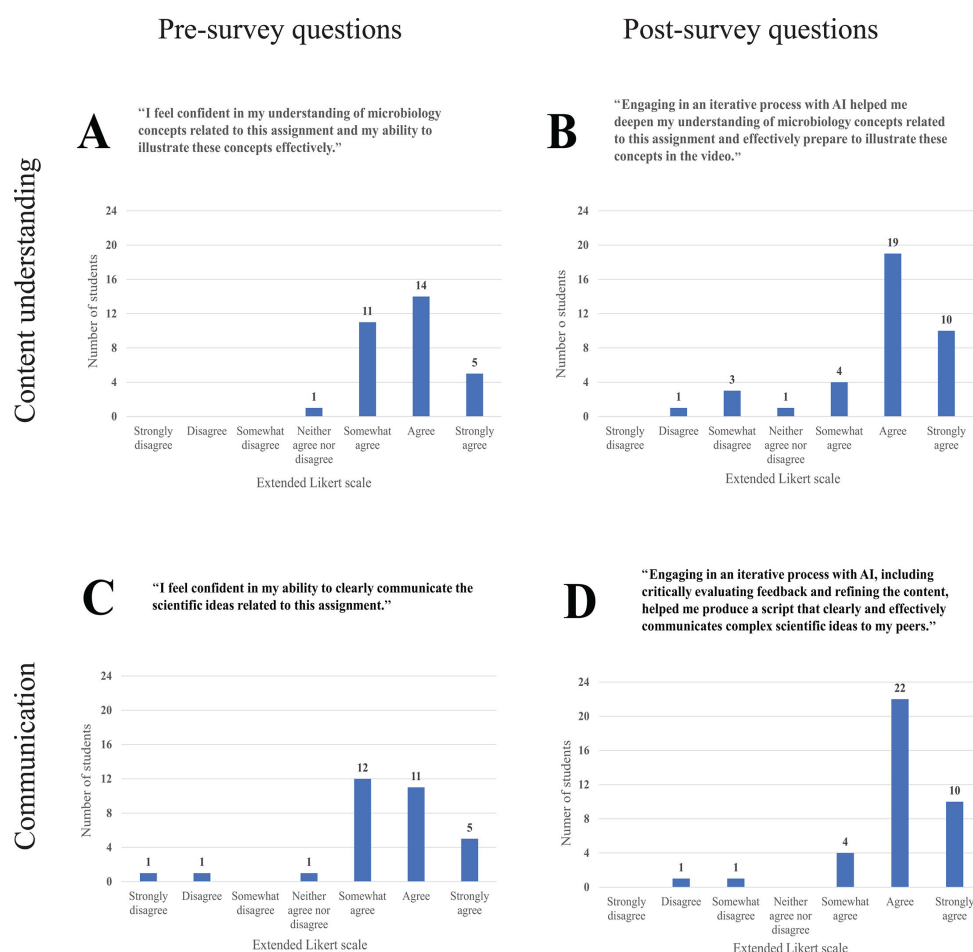


FIG 2 Student survey responses pertaining to assignment content using an extended Likert scale. (A) A majority of pre-survey respondents felt confident in their ability to understand and illustrate topics associated with the assignment (median = 6, mode = 6, and mean = 5.6). (B) Many post-survey respondents indicated that iterative engagement with the AI reinforced their understanding of the concepts related to the assignment (median = 6, mode = 6, and mean = 5.8). (C) Compared to their content understanding, pre-survey respondents felt less confident in their ability to communicate the information related to the assignment (median = 5, mode = 5, and mean = 5.3). (D) Many post-survey respondents felt more prepared to communicate the topic in their script after interacting with AI (median = 6, mode = 6, and mean = 6). In total, 48 students were enrolled in the course; 31 students responded to the pre-survey questions and 38 students responded to the post-survey questions. Faculty sought to incentivize student participation in the voluntary and anonymous survey by offering 1 extra credit point if 90% of the students across all sections participated in the survey.

Pre-survey questions

Post-survey questions

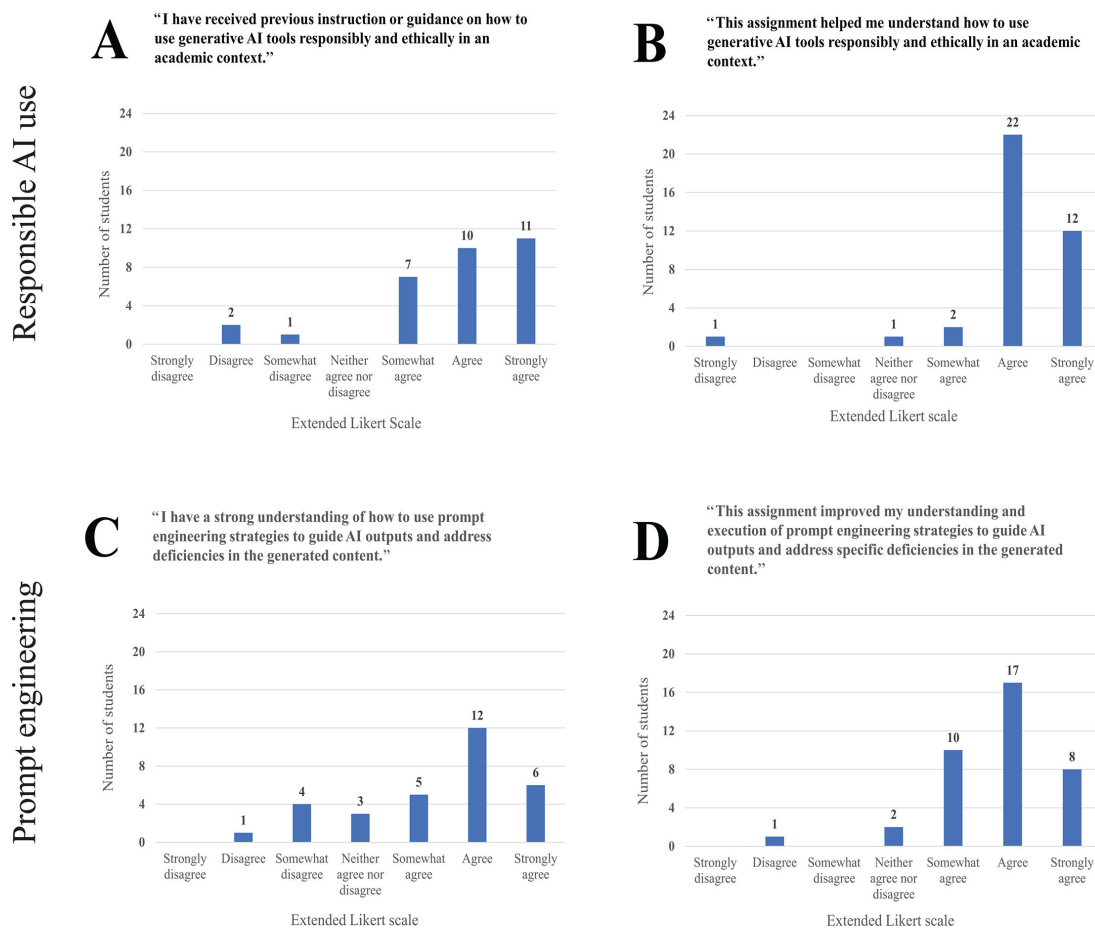


FIG 3 Student survey responses pertaining to AI literacy using an extended Likert scale. (A) Most pre-survey respondents reported having received prior instruction on the responsible use of AI (median = 6, mode = 6, and mean = 5.6). (B) Post-survey data suggest that engaging in the assignment helped reiterate previous instruction of responsible AI use (median = 6, mode = 6, and mean = 6.1). (C) Many pre-survey respondents felt confident in their prompt engineering skills prior to the start of the assignment (median = 6, mode = 6, and mean = 5.1). (D) Some post-survey respondents perceived an improved ability to prompt engineer after completing the assignment (median = 6, mode = 6, and mean = 5.7). In total, 48 students were enrolled in the course; 31 students responded to the pre-survey questions; and 38 students responded to the post-survey questions. Faculty sought to incentivize student participation in the voluntary and anonymous survey by offering one extra credit point if 90% of the students across all sections participated in the survey.

addressed the importance of prompt engineering and evaluation of AI outputs (Fig. 4; Appendix 7).

Self-reported student feedback suggests that iterative engagement with AI reinforced their conceptual grasp of material, improved their critical evaluation of feedback and refinement of script content, and increased confidence in AI literacy. This assignment is the result of extensive iterative refinement on the part of the authors and will necessarily continue to evolve based on student feedback and instructional observations. We plan to integrate summative assessment to track student understanding over time as well as incorporate greater guidance for students on how to evaluate AI outputs. We believe this writing assignment allows for the integration of AI and at the same time enables students to develop not only AI literacy but also critical thinking, self-reflection, and the practice of science communication. These skills are essential for students to remain academically and professionally competitive as AI tools continue to evolve.

Self-reflection question

Reflect on this iterative process and comment on how it impacted your learning compared to a more traditional assignment format. This reflection should be a minimum of 5 sentences.

Selected student responses

"This assignment helped me understand how learning can be active and layered instead of just writing a regular paper. Being able to revise my script multiple times showed me how to identify what information was missing and how I can make my script better. Working with ChatGPT helped me think carefully and made me realize the prompt you give them matters and how important it is to go into detail so the response will be clearer and more accurate. I liked that this process matched my learning style because I learn best when I can see progress and actively fix any mistakes. Overall, this whole experience was nice and educational. The only part that felt challenging was managing all the steps, but overall, I feel like I gained a better understanding of the content and improved at explaining complex microbiology topics in a simple and more engaging manner."

"I feel like a more traditional assignment format which forbids you from using AI is not a realistic way to approach and prepare us for the real world. We need to learn to adapt and use tools that are given to us properly. It helped me understand that AI is not perfect and needs a real person to overlook it. What aligned well with my learning style in this assignment is the CLARITY framework. It helped me organize my thoughts and everything necessary for this assignment. I like to gather my information before I begin something, and this helps me. What I felt was redundant was the REACT quick log. It confused me a little bit since there were multiple logs and they were all connected to each other. I am not sure if they were necessary."

"This process of forming multiple drafts was helpful to my learning, as it pointed out areas that I did not completely understand going into the drafting process. It allowed me to research independently and use the findings to better direct the AI to make a more complete version, and how to highlight the differences between the different ways that antibiotic resistance is passed to other cells. This assignment aligned well with my learning style as I tend to struggle with the overall structure of projects, and by having an AI come up with the sequences, it helps to give me an idea of how the presentation should flow. One thing I do not like about this process is that the AI is unable to provide good diagrams on mechanisms, so using the AI to do the research of how specific antibiotic resistance genes function would not be as enlightening as research articles on the topic."

FIG 4 Selected student responses. Three student answers in response to the final self-reflection question posed at the end of Part 1 of the assignment.

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ETHICS APPROVAL

This study was submitted to the Thomas Jefferson University Institutional Review Board and was determined to be exempt (IRISID-2025-0095).

ADDITIONAL FILES

The following material is available [online](#).

Supplemental Material

Supplemental material (jmbe00337-25-S0001.docx). Appendices 1 to 7.

REFERENCES

- Farazouli A, Cerratto-Pargman T, Bolander-Laksov K, McGrath C. 2024. Hello GPT! goodbye home examination? an exploratory study of AI chatbots impact on university teachers' assessment practices. *Assessment & Evaluation in Higher Education* 49:363–375. <https://doi.org/10.1080/02602938.2023.2241676>
- Fenton A. 2025. Reconsidering the use of oral exams and assessments: an old way to move into a new future. *Educ Res* 54:430–436. <https://doi.org/10.3102/0013189X251333638>
- Nikolic S, Daniel S, Haque R, Belkina M, Hassan GM, Grundy S, Lyden S, Neal P, Sandison C. 2023. ChatGPT versus engineering education assessment: a multidisciplinary and multi-institutional benchmarking and analysis of this generative artificial intelligence tool to investigate assessment integrity. *European Journal of Engineering Education* 48:559–614. <https://doi.org/10.1080/03043797.2023.2213169>
- Dawson P, Bearman M, Dollinger M, Boud D. 2024. Validity matters more than cheating. *Assessment & Evaluation in Higher Education* 49:1005–1016. <https://doi.org/10.1080/02602938.2024.2386662>
- Batista J, Mesquita A, Carnaz G. 2024. Generative AI and higher education: trends, challenges, and future directions from a systematic literature review. *Information* 15:676. <https://doi.org/10.3390/info15110676>
- Kasneci E, Sessler K, Küchemann S, Bannert M, Dementieva D, Fischer F, Gasser U, Groh G, Günemann S, Hüllermeier E, Krusche S, Kutyniok G, Michaeli T, Nerdel C, Pfeffer J, Poquet O, Sailer M, Schmidt A, Seidel T, Stadler M, Weller J, Kuhn J, Kasneci G. 2023. ChatGPT for good? on opportunities and challenges of large language models for education. *Learn Individ Differ* 103:102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Michel-Villarreal R, Vilalta-Perdomo E, Salinas-Navarro DE, Thierry-Aguilera R, Gerardou FS. 2023. Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences* 13:856. <https://doi.org/10.3390/educsci13090856>
- Abbas M, Jam FA, Khan TI. 2024. Is it harmful or helpful? examining the causes and consequences of generative AI usage among university students. *Int J Educ Technol High Educ* 21:10. <https://doi.org/10.1186/s41239-024-00444-7>
- Ivanov S. 2023. The dark side of artificial intelligence in higher education. *The Service Industries Journal* 43:1055–1082. <https://doi.org/10.1080/02642069.2023.2258799>
- Shimizu I, Kasai H, Shikino K, Araki N, Takahashi Z, Onodera M, Kimura Y, Tsukamoto T, Yamauchi K, Asahina M, Ito S, Kawakami E. 2023. Developing medical education curriculum reform strategies to address the impact of generative AI: qualitative study. *JMIR Med Educ* 9:e53466. <https://doi.org/10.2196/53466>
- Watermeyer R, Phipps L, Lanclos D, Knight C. 2024. Generative AI and the automating of academia. *Postdigit Sci Educ* 6:446–466. <https://doi.org/10.1007/s42438-023-00440-6>
- Covington EW, Kyle J, Moseley L. 2025. Scaffolding a clinical debate assignment in an infectious disease pharmacy elective: a mixed methods analysis. *Curr Pharm Teach Learn* 17:102306. <https://doi.org/10.1016/j.cptl.2025.102306>
- Dawn S, Dominguez KD, Troutman WG, Bond R, Cone C. 2011. Instructional scaffolding to improve students' skills in evaluating clinical literature. *Am J Pharm Educ* 75:62. <https://doi.org/10.5688/ajpe75462>
- Knoth N, Tolzin A, Janson A, Leimeister JM. 2024. AI literacy and its implications for prompt engineering strategies. *Computers and Education: Artificial Intelligence* 6:100225. <https://doi.org/10.1016/j.caeai.2024.100225>
- Farrokhnia M, Banihashem SK, Noroozi O, Wals A. 2024. A SWOT analysis of ChatGPT: implications for educational practice and research. *Innovations in Education and Teaching International* 61:460–474. <https://doi.org/10.1080/14703297.2023.2195846>
- Moustaghfir S, Brigui H. 2024. Navigating critical thinking in the digital era: an informative exploration. *IJLLT* 7:137–143. <https://doi.org/10.32996/ijllt.2024.7.1.11x>
- Lew MDN, Schmidt HG. 2011. Self-reflection and academic performance: is there a relationship? *Adv Health Sci Educ Theory Pract* 16:529–545. <https://doi.org/10.1007/s10459-011-9298-z>
- Ratnayake A, Bansal A, Wong N, Saseetharan T, Prompiengchai S, Jenne A, Thiagavel J, Ashok A. 2024. All “wrapped” up in reflection: supporting metacognitive awareness to promote students' self-regulated learning. *J Microbiol Biol Educ* 25:e00103-23. <https://doi.org/10.1128/jmbe.00103-23>
- Veine S, Anderson MK, Andersen NH, Espenes TC, Søyland TB, Wallin P, Reams J. 2020. Reflection as a core student learning activity in higher education - insights from nearly two decades of academic development. *International Journal for Academic Development* 25:147–161. <https://doi.org/10.1080/1360144X.2019.1659797>
- Fan JE, Bainbridge WA, Chamberlain R, Wammes JD. 2023. Drawing as a versatile cognitive tool. *Nat Rev Psychol* 2:556–568. <https://doi.org/10.1038/s44159-023-00212-w>
- Heideman PD, Flores KA, Sevier LM, Trouton KE. 2017. Effectiveness and adoption of a drawing-to-learn study tool for recall and problem solving: minute sketches with folded lists. *LSE* 16:ar28. <https://doi.org/10.1187/cbe.16-03-0116>
- Hutchins JA. 2020. Tailoring scientific communications for audience and research narrative. *CP Essential Lab Tech* 20:e40. <https://doi.org/10.1002/cpet.40>
- Ali M. 2025. The role of visual communication in education: Enhancing understanding and engagement through images and media. *International Conference “Actual Economy: Local Solutions for Global Challenges”*, p 253–255
- Hidayah LR. 2025. The importance of using visual in delivering information. *VCD* 8:52–61. <https://doi.org/10.37715/vcd.v8i1.2720>